

ENV 797 - Time Series Analysis for Energy and Environment Applications | Spring 2026

Assignment 3 - Due date 02/03/26

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Directions

You should open the .rmd file corresponding to this assignment on RStudio. The file is available on our class repository on Github.

Once you have the file open on your local machine the first thing you will do is rename the file such that it includes your first and last name (e.g., “LuanaLima_TSA_A03_Sp26.Rmd”). Then change “Student Name” on line 4 with your name.

Then you will start working through the assignment by **creating code and output** that answer each question. Be sure to use this assignment document. Your report should contain the answer to each question and any plots/tables you obtained (when applicable).

Please keep this R code chunk options for the report. It is easier for us to grade when we can see code and output together. And the tidy.opts will make sure that line breaks on your code chunks are automatically added for better visualization.

When you have completed the assignment, **Knit** the text and code into a single PDF file. Submit this pdf using Sakai.

Questions

Consider the same data you used for A2 from the spreadsheet “Table_10.1_Renewable_Energy_Production_and_Consumption”. The data comes from the US Energy Information and Administration and corresponds to the December 2025 Monthly Energy Review. This time you will work only with the following columns: **Total Renewable Energy Production**; and **Hydroelectric Power Consumption**.

Create a data frame structure with these two time series only.

R packages needed for this assignment: “forecast”, “tseries”, and “Kendall”. Install these packages, if you haven’t done yet. Do not forget to load them before running your script, since they are NOT default packages.\

```
#Load/install required package
```

```
library(tseries)
library(Kendall)
library(dplyr)
library(cowplot)
library(readxl)
library(lubridate)
library(forecast)
library(ggplot2)
```

```
##Trend Component
```

Q1

For each series (Total Renewable Production and Hydroelectric Consumption) create three plots arranged in a row (side-by-side): (1) time series plot, (2) ACF, (3) PACF. Use `cowplot::plot_grid()` to place them in a grid.

```
url <- "https://www.eia.gov/totalenergy/data/browser/xls.php?tbl=T10.01&freq=m"
tmp <- tempfile(fileext = ".xlsx")
download.file(url, tmp, mode = "wb")

eia_excel_energy <- read_excel(tmp, skip = 12, col_names = FALSE)
```

```
## New names:
## * ' -> '...1'
## * ' -> '...2'
## * ' -> '...3'
## * ' -> '...4'
## * ' -> '...5'
## * ' -> '...6'
## * ' -> '...7'
## * ' -> '...8'
## * ' -> '...9'
## * ' -> '...10'
## * ' -> '...11'
## * ' -> '...12'
## * ' -> '...13'
## * ' -> '...14'
```

```
read_col_names <- read_excel(tmp, skip = 10, n_max = 1, col_names = FALSE)
```

```
## New names:
## * ' -> '...1'
## * ' -> '...2'
## * ' -> '...3'
## * ' -> '...4'
## * ' -> '...5'
## * ' -> '...6'
## * ' -> '...7'
## * ' -> '...8'
## * ' -> '...9'
## * ' -> '...10'
## * ' -> '...11'
## * ' -> '...12'
## * ' -> '...13'
## * ' -> '...14'
```

```
colnames(eia_excel_energy) <- as.character(unlist(read_col_names))
```

```
#Select total renewable energy and hydroelectric
energy_subset <- eia_excel_energy %>%
  select(`Total Renewable Energy Production`,
         `Hydroelectric Power Consumption`) %>%
```

```
mutate(across(everything(), as.numeric))

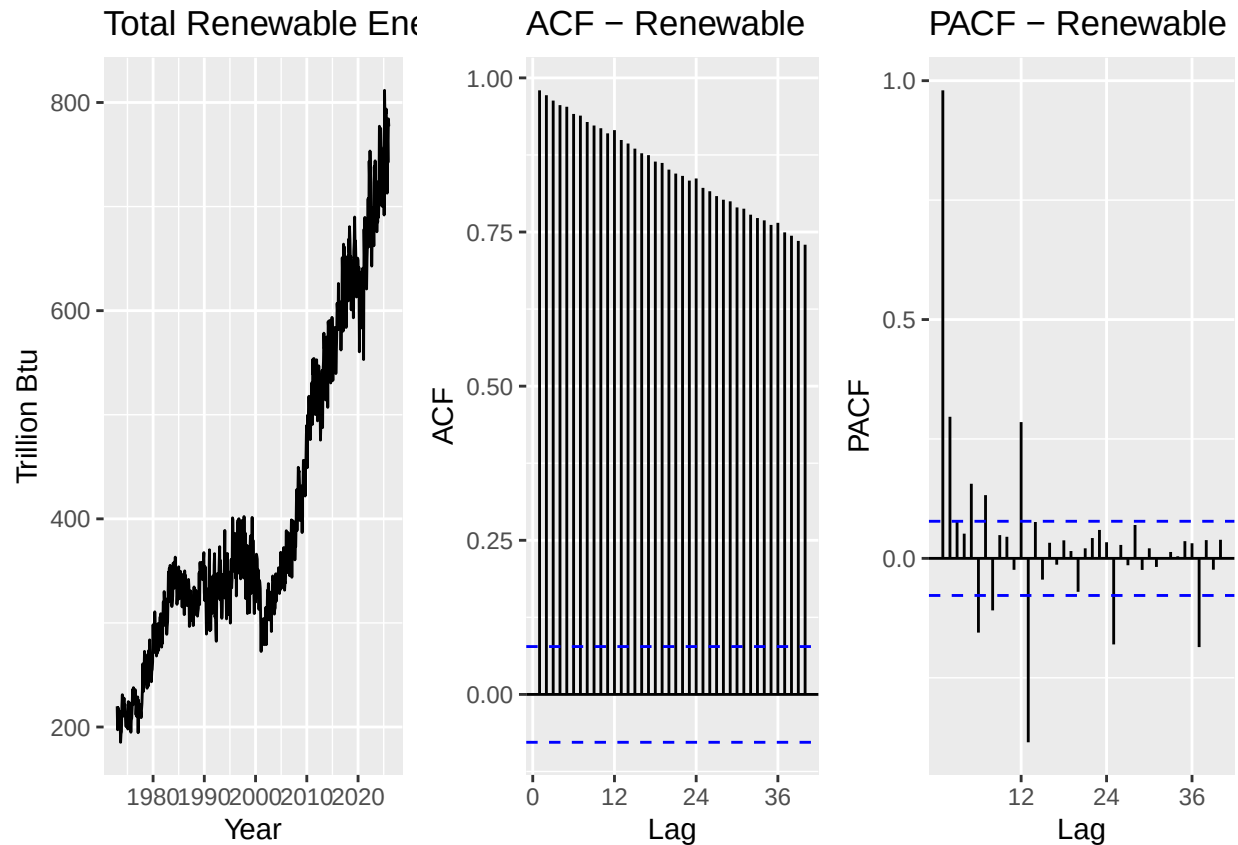
#Convert to time series object
energy_ts <- ts(energy_subset, start = c(1973, 1), frequency = 12)
nobserved <- nrow(energy_subset)

head(energy_ts)
```

```
##           Total Renewable Energy Production Hydroelectric Power Consumption
## Jan 1973                                219.839                        89.562
## Feb 1973                                197.330                        79.544
## Mar 1973                                218.686                        88.284
## Apr 1973                                209.330                        83.152
## May 1973                                215.982                        85.643
## Jun 1973                                208.249                        82.060
```

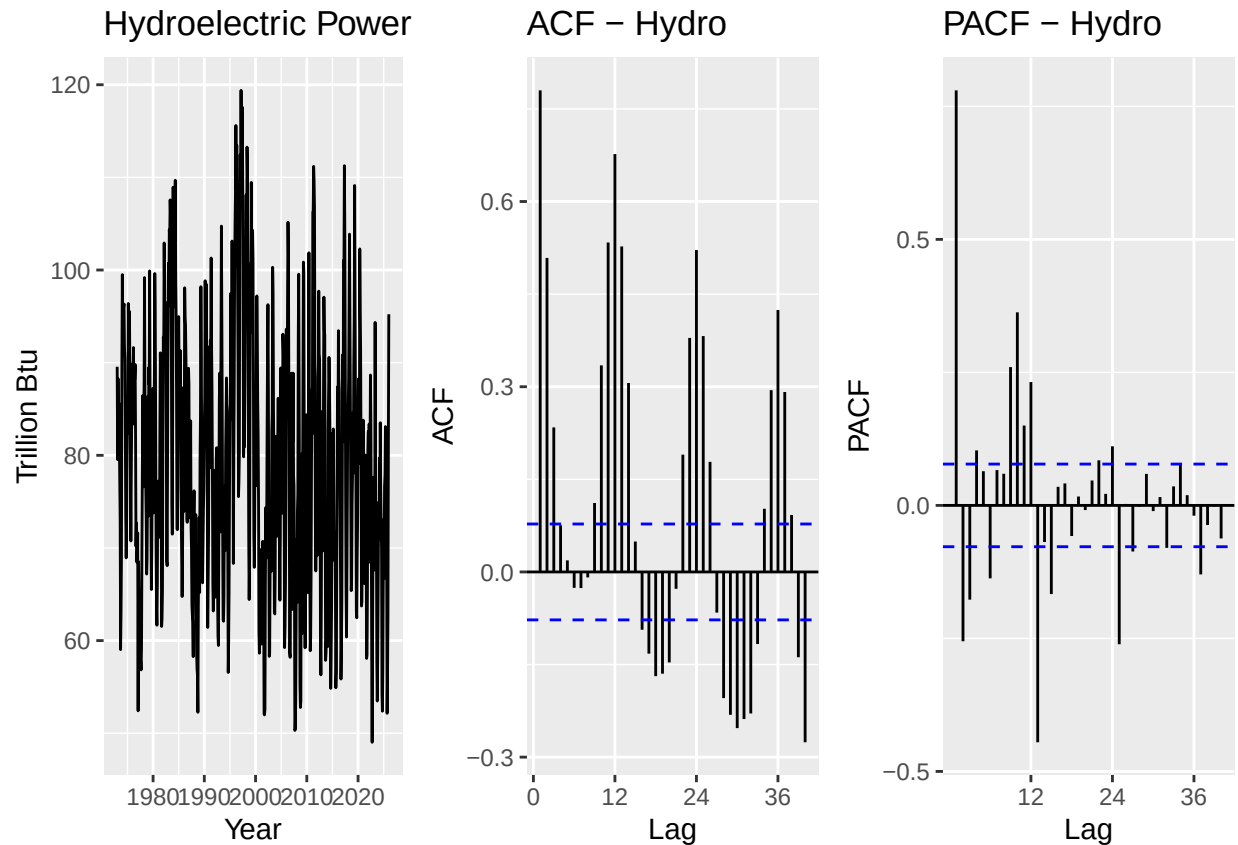
```
plot_grid(
  autoplot(energy_ts[, "Total Renewable Energy Production"]) +
    labs(title = "Total Renewable Energy Production",
         x = "Year", y = "Trillion Btu"),
  autoplot(Acf(energy_ts[, "Total Renewable Energy Production"],
               lag.max = 40, plot = FALSE),
            main = "ACF - Renewable"),
  autoplot(Pacf(energy_ts[, "Total Renewable Energy Production"],
                lag.max = 40, plot = FALSE),
            main = "PACF - Renewable"),
  nrow = 1
)
```

```
## Warning in ggplot2::geom_segment(lineend = "butt", ...): Ignoring unknown parameters: 'main'
## Ignoring unknown parameters: 'main'
```



```
#Plot grid for Hydroelectric Power Consumption
plot_grid(
  autoplot(energy_ts[, "Hydroelectric Power Consumption"]) +
    labs(title = "Hydroelectric Power Consumption",
         x = "Year", y = "Trillion Btu"),
  autoplot(Acf(energy_ts[, "Hydroelectric Power Consumption"],
               lag.max = 40, plot = FALSE),
            main = "ACF - Hydro"),
  autoplot(Pacf(energy_ts[, "Hydroelectric Power Consumption"],
                lag.max = 40, plot = FALSE),
            main = "PACF - Hydro"),
  nrow = 1
)
```

```
## Warning in ggplot2::geom_segment(lineend = "butt", ...): Ignoring unknown parameters: 'main'
## Ignoring unknown parameters: 'main'
```



Q2

From the plot in Q1, do the series Total Renewable Energy Production and Hydroelectric Power Consumption appear to have a trend? If yes, what kind of trend?

Answer: Total Renewable Energy Production has a clear upward trend, going from around 200 in 1973 to over 800 by 2025. The slow decay in the ACF also confirms the trend. Hydroelectric Power Consumption does not show a clear long-term trend, it stays around the same range over time but shows clear seasonal cycles in both the time series and the ACF plot.

Q3

Use the `lm()` function to fit a linear trend to the two time series. Ask R to print the summary of the regression. Interpret the regression output, i.e., slope and intercept. Save the regression coefficients for further analysis.

```
#Create time vector
t <- c(1:nobserved)

#Linear trend for Total Renewable Energy Production
lm_renewable <- lm(energy_subset$`Total Renewable Energy Production` ~ t)
summary(lm_renewable)
```

##

```
## Call:
## lm(formula = energy_subset$`Total Renewable Energy Production` ~
##     t)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -155.76  -39.66   13.49   41.98  167.15
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 169.98733     5.13162   33.12  <2e-16 ***
## t           0.75687     0.01394   54.31  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 64.68 on 635 degrees of freedom
## Multiple R-squared:  0.8228, Adjusted R-squared:  0.8226
## F-statistic: 2949 on 1 and 635 DF,  p-value: < 2.2e-16

#Linear trend for Hydroelectric Power Consumption
lm_hydro <- lm(energy_subset$`Hydroelectric Power Consumption` ~ t)
summary(lm_hydro)
```

```
##
## Call:
## lm(formula = energy_subset$`Hydroelectric Power Consumption` ~
##     t)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -30.202 -10.294  -0.686    8.921   39.729
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 83.240024     1.107822   75.14  < 2e-16 ***
## t          -0.012277     0.003009   -4.08 5.07e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.96 on 635 degrees of freedom
## Multiple R-squared:  0.02555, Adjusted R-squared:  0.02402
## F-statistic: 16.65 on 1 and 635 DF,  p-value: 5.069e-05
```

```
#Save coefficients for Renewable
beta0_renewable <- as.numeric(lm_renewable$coefficients[1])
beta1_renewable <- as.numeric(lm_renewable$coefficients[2])

#Save coefficients for Hydro
beta0_hydro <- as.numeric(lm_hydro$coefficients[1])
beta1_hydro <- as.numeric(lm_hydro$coefficients[2])
```

Answer: For Total Renewable Energy Production, the intercept (169.99) is the estimated value at the start of the series and the slope (0.757) means production grows by about 0.76 Trillion

Btu each month on average. The slope is highly significant $p < 2e-16$, which confirms the strong upward trend. For Hydroelectric Power Consumption, the intercept is 83.24 and the slope is -0.012, meaning hydro consumption drops by a tiny amount each month. The slope is statistically significant but very small, so the trend is barely there compared to the renewable series.

Q4

Use the regression coefficients to detrend each series (subtract fitted linear trend). Plot detrended series and compare with the original time series from Q1. Describe what changed.

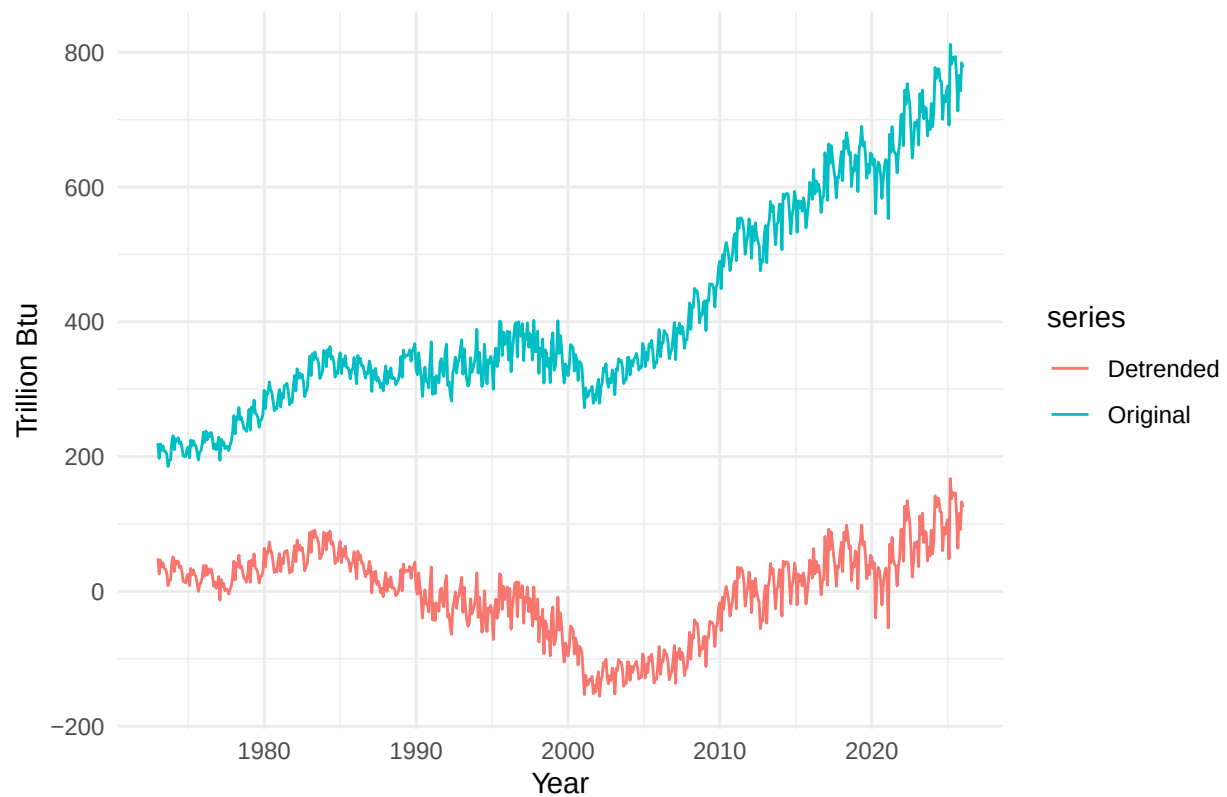
```
#Detrend Renewable
detrend_renewable <- energy_subset$`Total Renewable Energy Production` -
  (beta0_renewable + beta1_renewable * t)

#Detrend Hydro
detrend_hydro <- energy_subset$`Hydroelectric Power Consumption` -
  (beta0_hydro + beta1_hydro * t)

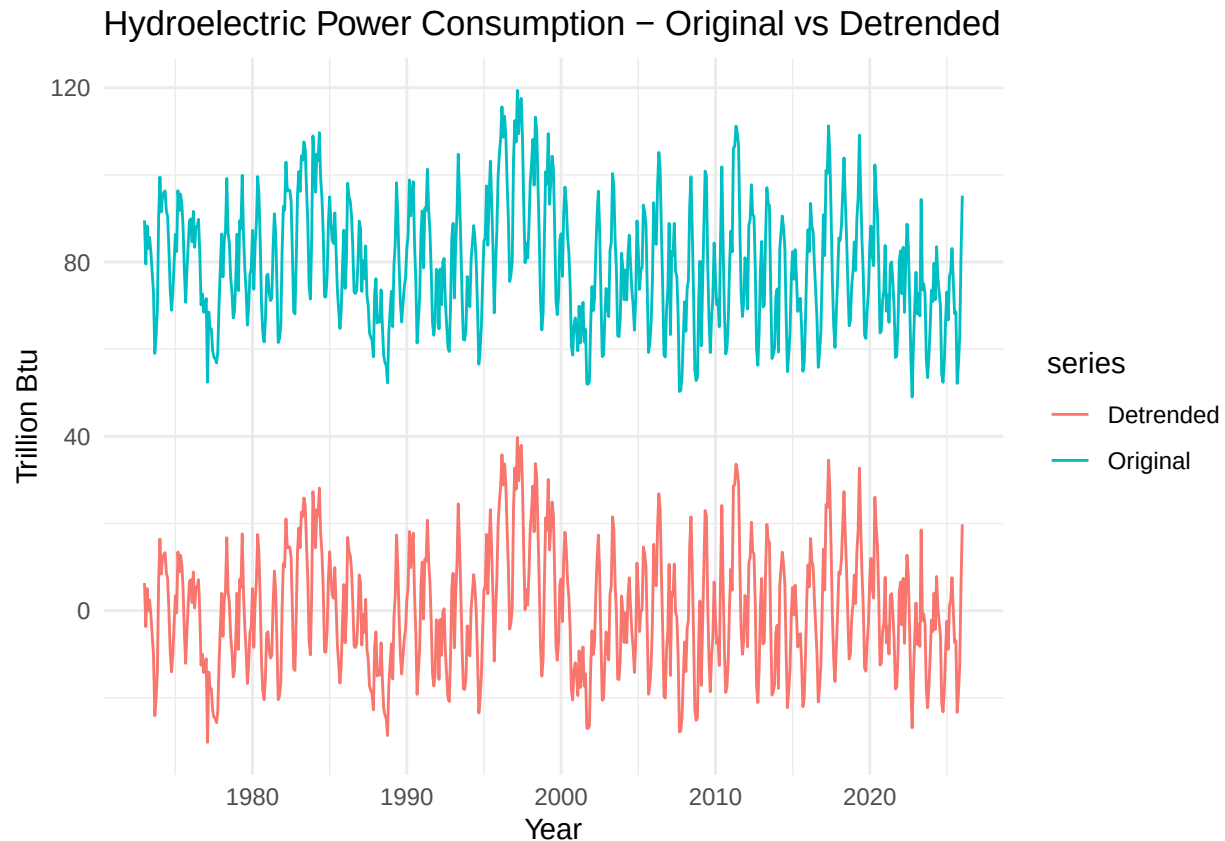
#Convert to ts objects
ts_detrend_renewable <- ts(detrend_renewable, start = c(1973, 1), frequency = 12)
ts_detrend_hydro <- ts(detrend_hydro, start = c(1973, 1), frequency = 12)

#Compare original vs detrended for Renewable
autoplot(energy_ts[, "Total Renewable Energy Production"], series = "Original") +
  autolayer(ts_detrend_renewable, series = "Detrended") +
  labs(title = "Total Renewable Energy Production - Original vs Detrended",
       x = "Year", y = "Trillion Btu") +
  theme_minimal()
```

Total Renewable Energy Production – Original vs Detrended



```
#Compare original vs detrended for Hydro
autoplot(energy_ts[, "Hydroelectric Power Consumption"], series = "Original") +
  autolayer(ts_detrend_hydro, series = "Detrended") +
  labs(title = "Hydroelectric Power Consumption - Original vs Detrended",
        x = "Year", y = "Trillion Btu") +
  theme_minimal()
```

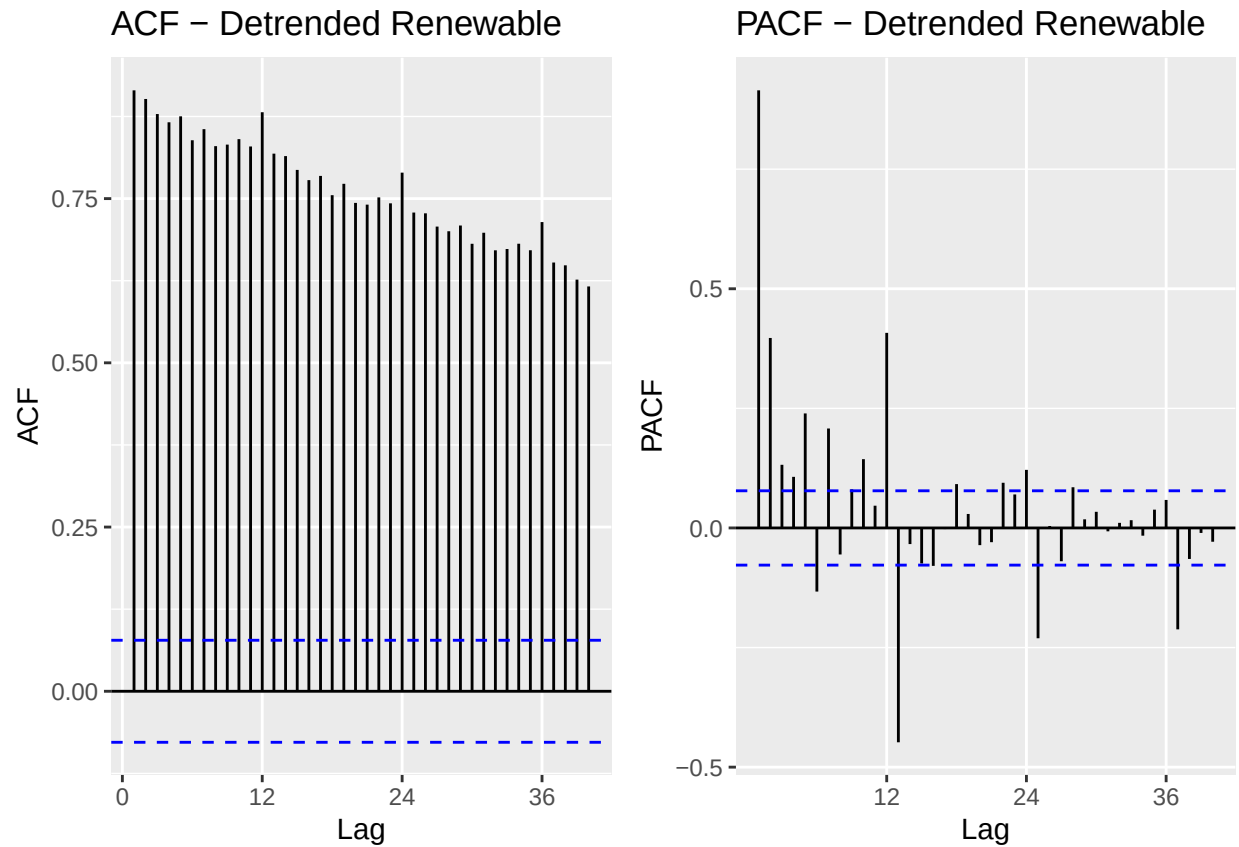



Answer: For Total Renewable Energy Production, the detrended series no longer climbs from 200 to 800, instead it fluctuates around zero. The upward trend was removed by subtracting the fitted line. For Hydroelectric Power Consumption, the detrended series looks almost the same shape as the original but shifted down to fluctuate around zero, because there was barely any trend to remove.

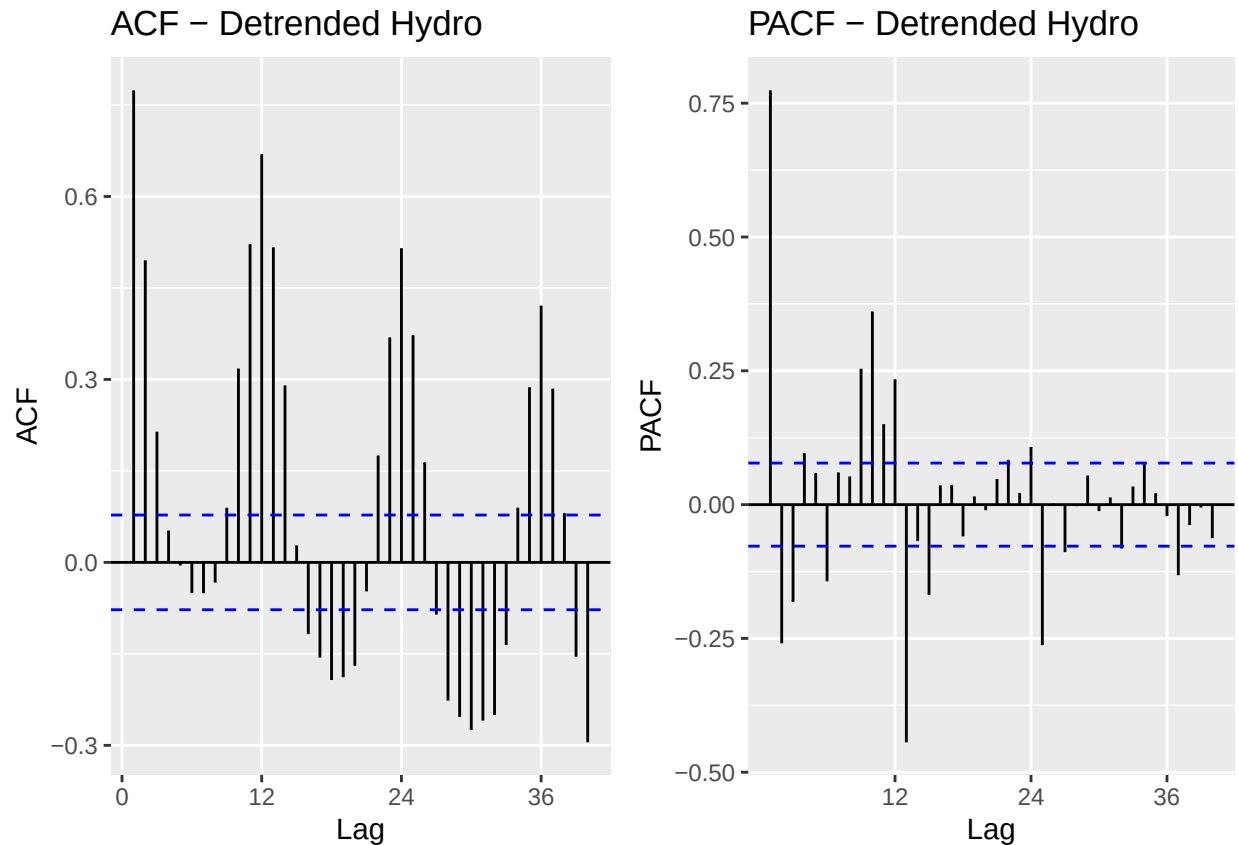
Q5

Plot ACF and PACF for the detrended series and compare with the plots from Q1. You may use `plot_grid()` again to get them side by side to make it easier to compare. Did the plots change? How?

```
#ACF and PACF for detrended Renewable
plot_grid(
  autoplot(Acf(ts_detrend_renewable, lag.max = 40, plot = FALSE),
    main = "ACF - Detrended Renewable"),
  autoplot(Pacf(ts_detrend_renewable, lag.max = 40, plot = FALSE),
    main = "PACF - Detrended Renewable"),
  nrow = 1
)
```



```
#ACF and PACF for detrended Hydro
plot_grid(
  autoplot(Acf(ts_detrend_hydro, lag.max = 40, plot = FALSE),
    main = "ACF - Detrended Hydro"),
  autoplot(Pacf(ts_detrend_hydro, lag.max = 40, plot = FALSE),
    main = "PACF - Detrended Hydro"),
  nrow = 1
)
```



Answer: For Total Renewable Energy Production, the ACF still shows slow decay but it comes down faster than in Q1, and the seasonal peaks at lags 12, 24, and 36 are now more visible. This shows the linear detrending helped but did not fully remove the autocorrelation. For Hydroelectric Power Consumption, the ACF and PACF look nearly the same as Q1 because there was no real trend to remove.

Seasonal Component

Set aside the detrended series and consider the original series again from Q1 to answer Q6 to Q8.

Q6

Just by looking at the time series and the acf plots, do the series seem to have a seasonal trend? No need to run any code to answer your question. Just type in your answer below.

Answer: Hydroelectric Power Consumption clearly has a seasonal pattern, the time series shows repeating yearly cycles and the ACF has strong peaks at lags 12, 24, and 36. Total Renewable Energy Production does not show clear seasonality, the trend dominates the plot.

Q7

Use function `lm()` to fit a seasonal means model (i.e. using the seasonal dummies) to the two time series. Ask R to print the summary of the regression. Interpret the regression output. From the results, which series have a seasonal trend? Do the results match your answer to Q6?

```
#Seasonal dummies
dummies <- seasonaldummy(energy_ts[, "Total Renewable Energy Production"])

#Seasonal means model for Renewable (using ORIGINAL series, not detrended)
seas_renewable <- lm(energy_subset$`Total Renewable Energy Production` ~ dummies)
summary(seas_renewable)
```

```
##
## Call:
## lm(formula = energy_subset$`Total Renewable Energy Production` ~
##     dummies)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -213.32  -99.26  -62.58   125.54   388.50
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  424.1913    21.1899   20.019  <2e-16 ***
## dummiesJan     1.7671    29.8280    0.059   0.953
## dummiesFeb   -41.4766    29.9670   -1.384   0.167
## dummiesMar    -0.9916    29.9670   -0.033   0.974
## dummiesApr   -13.8431    29.9670   -0.462   0.644
## dummiesMay     1.2148    29.9670    0.041   0.968
## dummiesJun    -9.1664    29.9670   -0.306   0.760
## dummiesJul    -3.0727    29.9670   -0.103   0.918
## dummiesAug   -10.9013    29.9670   -0.364   0.716
## dummiesSep   -35.9986    29.9670   -1.201   0.230
## dummiesOct   -19.9182    29.9670   -0.665   0.507
## dummiesNov   -21.0190    29.9670   -0.701   0.483
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 154.3 on 625 degrees of freedom
## Multiple R-squared:  0.008154, Adjusted R-squared: -0.009302
## F-statistic: 0.4671 on 11 and 625 DF, p-value: 0.9235
```

```
#Seasonal means model for Hydro (using ORIGINAL series, not detrended)
seas_hydro <- lm(energy_subset$`Hydroelectric Power Consumption` ~ dummies)
summary(seas_hydro)
```

```
##
## Call:
## lm(formula = energy_subset$`Hydroelectric Power Consumption` ~
##     dummies)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -30.895  -6.452  -0.595    6.177   32.557
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept)    79.760      1.420   56.166 < 2e-16 ***
## dummiesJan      5.111      1.999    2.557 0.010801 *
## dummiesFeb     -2.451      2.008   -1.220 0.222792
## dummiesMar      7.080      2.008    3.525 0.000454 ***
## dummiesApr      5.578      2.008    2.778 0.005641 **
## dummiesMay     14.044      2.008    6.993 6.94e-12 ***
## dummiesJun     10.744      2.008    5.350 1.24e-07 ***
## dummiesJul      3.968      2.008    1.976 0.048637 *
## dummiesAug     -5.348      2.008   -2.663 0.007946 **
## dummiesSep     -16.636      2.008   -8.284 7.29e-16 ***
## dummiesOct     -16.496      2.008   -8.214 1.23e-15 ***
## dummiesNov     -10.933      2.008   -5.444 7.50e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 10.34 on 625 degrees of freedom
## Multiple R-squared:  0.4743, Adjusted R-squared:  0.465
## F-statistic: 51.25 on 11 and 625 DF,  p-value: < 2.2e-16
```

For Total Renewable Energy Production, none of the monthly coefficients are statistically significant and the R^2 is only 0.008, which means seasonality explains almost none of the variation. For Hydroelectric Power Consumption, most monthly coefficients are highly significant and the R^2 is 0.474, meaning seasonality explains nearly half of the variation. This matches my answer to Q6, only Hydro has a strong seasonal pattern.

Q8

Use the regression coefficients from Q7 to deseason the series. Plot the deseason series and compare with the plots from part Q1. Did anything change?

```
#Save coefficients for Renewable seasonal model
beta_int_renewable <- seas_renewable$coefficients[1]
beta_coeff_renewable <- seas_renewable$coefficients[2:12]

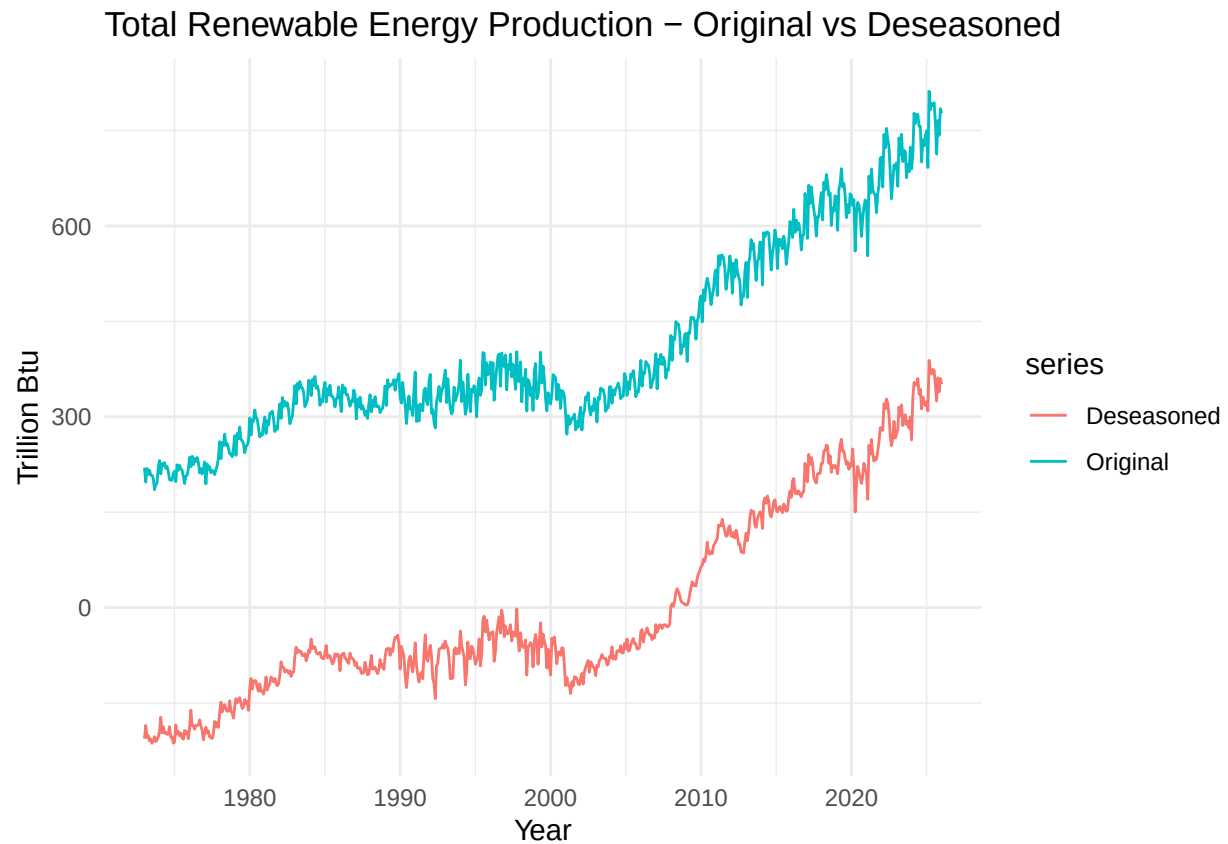
#Save coefficients for Hydro seasonal model
beta_int_hydro <- seas_hydro$coefficients[1]
beta_coeff_hydro <- seas_hydro$coefficients[2:12]

#Compute seasonal component for each observation
season_renewable <- array(0, nobserved)
season_hydro <- array(0, nobserved)
for(i in 1:nobserved){
  season_renewable[i] <- beta_int_renewable + beta_coeff_renewable %*% dummies[i,]
  season_hydro[i] <- beta_int_hydro + beta_coeff_hydro %*% dummies[i,]
}

#Subtract seasonal component from original
deseason_renewable <- energy_subset$`Total Renewable Energy Production` - season_renewable
deseason_hydro <- energy_subset$`Hydroelectric Power Consumption` - season_hydro

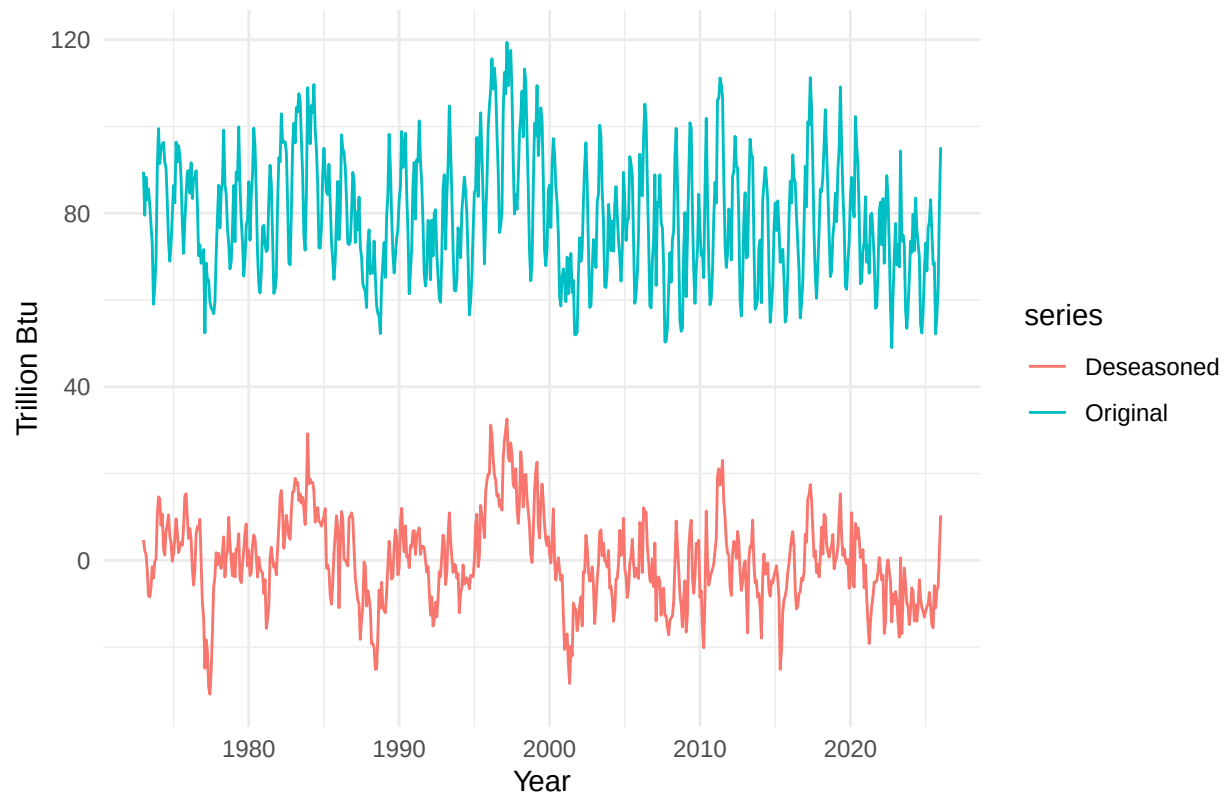
#Convert to ts objects
ts_deseason_renewable <- ts(deseason_renewable, start = c(1973, 1), frequency = 12)
ts_deseason_hydro <- ts(deseason_hydro, start = c(1973, 1), frequency = 12)
```

```
#Compare original vs deseasoned for Renewable
autoplot(energy_ts[, "Total Renewable Energy Production"], series = "Original") +
  autolayer(ts_deseason_renewable, series = "Deseasoned") +
  labs(title = "Total Renewable Energy Production - Original vs Deseasoned",
       x = "Year", y = "Trillion Btu") +
  theme_minimal()
```



```
#Compare original vs deseasoned for Hydro
autoplot(energy_ts[, "Hydroelectric Power Consumption"], series = "Original") +
  autolayer(ts_deseason_hydro, series = "Deseasoned") +
  labs(title = "Hydroelectric Power Consumption - Original vs Deseasoned",
       x = "Year", y = "Trillion Btu") +
  theme_minimal()
```

Hydroelectric Power Consumption – Original vs Deseasoned

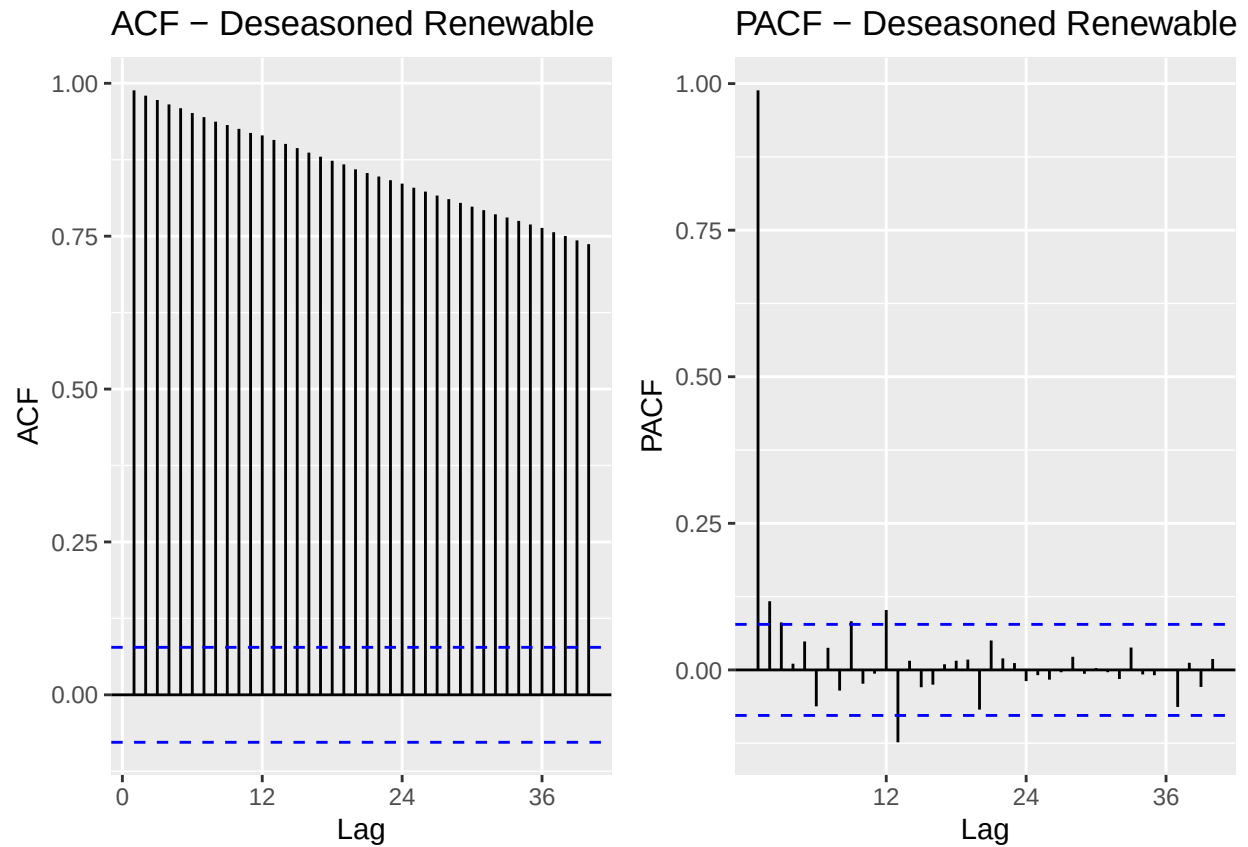


For Hydroelectric Power Consumption, the deseasoned series is much smoother than the original, the sharp yearly cycles are gone. For Total Renewable Energy Production, the deseasoned series looks almost identical to the original because there was barely any seasonality to remove, the upward trend is still there.

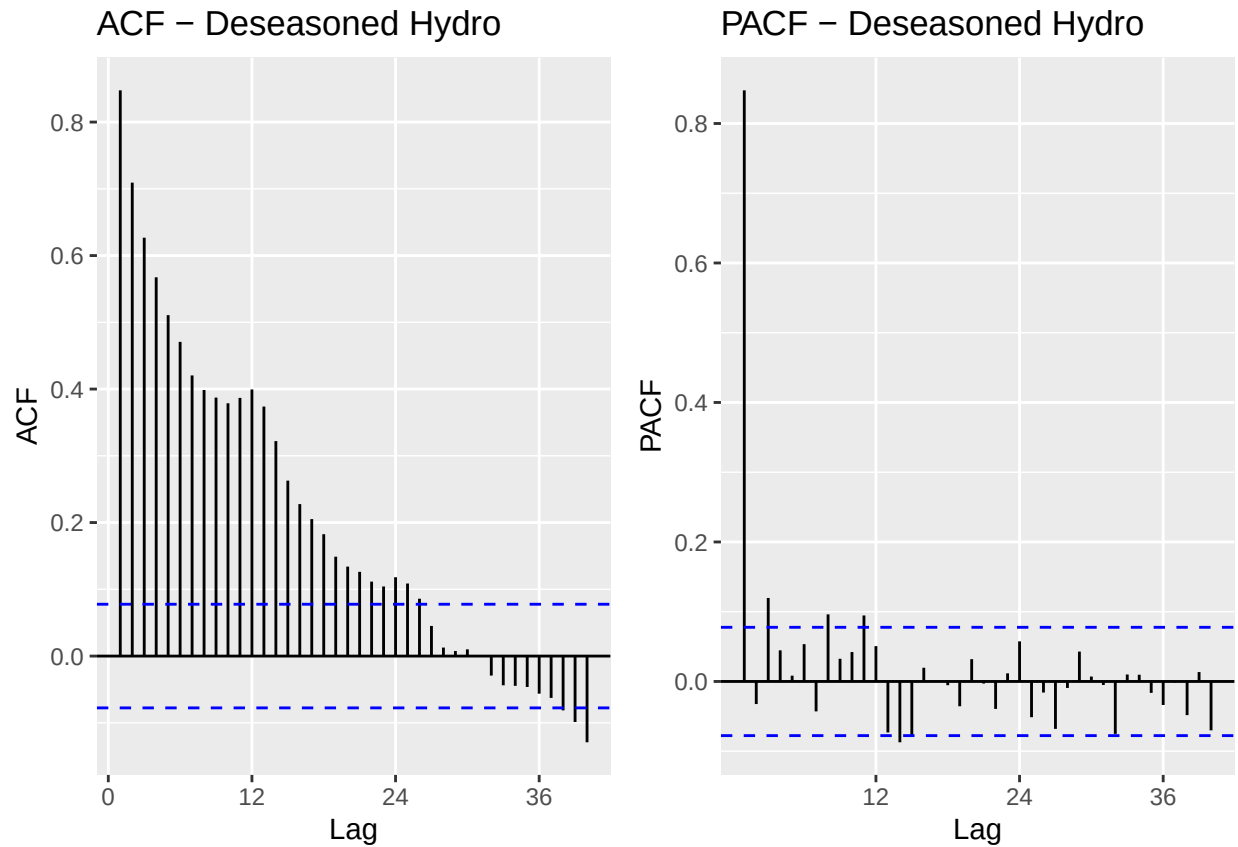
Q9

Plot ACF and PACF for the deseason series and compare with the plots from Q1. You may use `plot_grid()` again to get them side by side. Did the plots change? How?

```
#ACF and PACF for deseasoned Renewable
plot_grid(
  autoplot(Acf(ts_deseason_renewable, lag.max = 40, plot = FALSE),
    main = "ACF - Deseasoned Renewable"),
  autoplot(Pacf(ts_deseason_renewable, lag.max = 40, plot = FALSE),
    main = "PACF - Deseasoned Renewable"),
  nrow = 1
)
```



```
#ACF and PACF for deseasoned Hydro
plot_grid(
  autoplot(Acf(ts_deseason_hydro, lag.max = 40, plot = FALSE),
    main = "ACF - Deseasoned Hydro"),
  autoplot(Pacf(ts_deseason_hydro, lag.max = 40, plot = FALSE),
    main = "PACF - Deseasoned Hydro"),
  nrow = 1
)
```

Answer: For Hydroelectric Power Consumption, the seasonal peaks at lags 12, 24, and 36 are gone in the ACF, and what is left is a slow decay that looks like residual autocorrelation. For Total Renewable Energy Production, the ACF still shows the same slow decay as Q1 because the trend was not removed in this step, only seasonality was.